

The Low- p Vector-Valued Hardy-Space Extension Was Explicitly Answered by arXiv:2001.10021

Run `fa_banach_001`, lane 8
`agent_lane_08` (GPT5.6)

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Status: `literature_already_answered`. This is an exact two-paper identification and does not count as a new mathematical solution produced by the run.

1 Source problem

The source is R. D. Carmichael, S. Pilipović, and J. Vindas, *Vector valued Hardy spaces related to analytic functions having distributional boundary values*, arXiv:1801.06709 (published 2019) [1]. Theorems 4.2–4.4 establish Hardy-space membership and Poisson representation for Hilbert-valued analytic functions in the range $2 \leq p \leq \infty$. Section 5, PDF page 20, states:

For the cases $1 \leq p < 2$, results like Theorems 4.2, 4.3, and 4.4 require separate proof techniques. The authors propose to use Banach spaces having Fourier type p , and say that future research will obtain those theorems in the low- p range.

Thus the concrete target is to pass below $p = 2$, with Fourier type suggested as the additional Banach-space hypothesis.

2 Explicit supporting answer

The separate later paper by the same authors, *Note on vector valued Hardy spaces related to analytic functions having distributional boundary values*, arXiv:2001.10021 (2020) [2], explicitly identifies the source and its Theorem 4.4. Its introduction, PDF page 1, says that the earlier paper obtained the vector-valued result only for $2 \leq p \leq \infty$ and only for Hilbert targets, and then states that the aim of the note is to prove it for every $1 \leq p \leq \infty$ and every Banach space.

Theorem 1, on supporting PDF page 2, gives the promised statement. If X is any Banach space, C is a regular open convex cone, $p \in [1, \infty]$, and an X -valued holomorphic function $F : T^C \rightarrow X$ has distributional boundary value $f \in L^p(\mathbb{R}^n, X)$, then

$$F \in H^p(T^C, X), \quad \|F\|_{H^p} = \|f\|_{L^p},$$

and the proof identifies F with the Poisson integral of f .

3 Why this answers the proposal

Feature	2019 source	2020 answer
Exponent range	$2 \leq p \leq \infty$	$1 \leq p \leq \infty$
Target space	Hilbert space	arbitrary Banach space
Suggested extra input	Fourier type p	none
Conclusion	Hardy membership/Poisson form	same, in stronger scope

The later theorem therefore contains the proposed low- p extension as a strict special case. This relation is explicit rather than agent-inferred: the supporting authors cite the earlier paper, name Theorem 4.4, describe its exact limitation, and announce the improvement before proving it.

4 Scope and classification

No part of the source’s stated low- p extension remains open under the later theorem’s hypotheses. The correct run classification is `literature_already_answered`, not a new full or partial solution. The source and supporting PDFs are stored with this packet; the precise source locations and scope audit are recorded in `verification.md`.

References

- [1] R. D. Carmichael, S. Pilipović, and J. Vindas, *Vector valued Hardy spaces related to analytic functions having distributional boundary values*, Complex Variables and Elliptic Equations 64 (2019), 1634–1654; arXiv:1801.06709.
- [2] R. D. Carmichael, S. Pilipović, and J. Vindas, *Note on vector valued Hardy spaces related to analytic functions having distributional boundary values*, arXiv:2001.10021, 2020.